

Lorenzo Sarnataro

Curriculum Vitæ

Personal information

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Appointments

2024 – PRESENT **University of Toronto, Brauer Postdoctoral Fellow**
Mentors: Prof. Yevgeny Liokumovich and Prof. Robert Haslhofer.

Education

2017 – 2018 AND **Ph.D in Mathematics, Princeton University, USA**
2020 – 2024 *Advisor:* Prof. Fernando Codá Marques.
Thesis: Two regularity results in the theory of minimal hypersurfaces
2013 – 2017 **Master of Mathematics (MMath), University of Oxford, UK**
Distinction in Prelims (first year exams)
First class Honours in Part A and Part B (second and third year exams)
First class Honours in Part C (fourth year exams)

Publications and Preprints

- 2026 *On the Gage–Hamilton–Grayson Theorem for free boundary curves* (with Yueheng Bao) – preprint, 14 pages, (arXiv:2610.09557)
- 2026 *Regularity and compactness theory of stable free boundary minimal hypersurfaces* (with Costante Bellettini, Martin Li, and Davide Parise) – preprint, 61 pages, (arXiv:2610.09596)
- 2026 *Boundary behaviour of stable Allen–Cahn limit interfaces in Riemannian manifolds* (with Martin Li and Davide Parise) – preprint, 60 pages, (arXiv:2610.04833)
- 2026 *Second order estimates on transition layers for the Allen–Cahn equation with homogeneous Neumann boundary conditions: Part I – Boundary-orthogonal transition layers* (with Akashdeep Dey, Wenkui Du, and Davide Parise) – preprint, 107 pages (arXiv:2609.32829)
- 2026 *Existence of free boundary minimal disks in convex regions* (with Douglas Stryker, Zhichao Wang, and Xin Zhou) – preprint, 22 pages (arXiv:2606.02396) – submitted to **J. Eur. Math. Soc.**
- 2024 *Index, Intersections, and Multiplicity of Min-Max Geodesics* (with Jared Marx-Kuo and Douglas Stryker), **Advances in Mathematics**, Volume 502 (2026), Part C. Article 111193, pages 1–46, (arXiv:2410.02580)
- 2023 *Boundary behavior of limit-interfaces for the Allen–Cahn equation on Riemannian manifolds with Neumann boundary condition* (with Martin Li and Davide Parise), **Arch. Rational Mech. Anal.** 248, **124** (2024), pages 1–43, (arXiv:2312.07210)
- 2023 *Existence of closed embedded curves of constant curvature via min-max* (with Douglas Stryker), **J. Reine Angew. Math. (Crelles Journal)**, vol. 2026, no. 831, pages 185–211, (arXiv:2306.04840)
- 2023 *Optimal regularity for minimizers of the prescribed mean curvature functional over isotopies* (with Douglas Stryker), **Camb. J. Math.** **13** (2025), no. 3, pages 609–706, (arXiv:2304.02722)

Scholarships and Awards

2021-2024 *Bradford H. Arnold *42 and Mary Ellen Arnold Fellowship* (Princeton University)

- 2016 *Junior Mathematical Prize* (Mathematical Institute, University of Oxford) for combined performance in Part A and Part B
- 2014 - 2017 *Worcester College's Scholarship* for Distinction in Prelims, Part A and Part B
- 2014 *Fondazione M. Gaia Scholarship* for outstanding academic performance, worth €10,000 over five years

Teaching

University of Toronto:

- WINTER 2027 *Instructor* for MATA37 (*Calculus II for Mathematical Sciences*) at UTSC
- FALL 2025 *Instructor* for MATA31 (*Calculus I for Mathematical Sciences*) at UTSC
- WINTER 2024 *Instructor and course head* for MATA31 (*Calculus I for Mathematical Sciences*) at UTSC

Princeton University:

- SPRING 2023 *Teaching assistant* for MAT 355: *Introduction to Differential Geometry*
- FALL 2023 *Teaching assistant* for MAT 300: *Multivariable Analysis I*
- SPRING 2023 *Precept instructor* for MAT 175: *Mathematics for Economics and Life Sciences*
- FALL 2022 *Precept instructor* for MAT 175: *Mathematics for Economics and Life Sciences*
- SPRING 2022 *Precept instructor* for MAT 201: *Multivariable Calculus*
- FALL 2021 *Precept instructor* for MAT 201: *Multivariable Calculus*
- SPRING 2021 *Teaching assistant* for MAT 201: *Multivariable Calculus*
- FALL 2020 *Teaching assistant* for MAT 300: *Multivariable Analysis I*

Invited Talks

- NOVEMBER 2026 *Analysis and Geometric Analysis Seminar* at Cornell University
- SEPTEMBER 2026 *Geometric Analysis Seminar* at Columbia University
- JUNE 2026 *International geometric analysis conference in Milan II* at Bocconi University
- MAY 2026 *Analysis Seminar* at ETH
- FEBRUARY 2026 *Geometry Seminar* at Stanford University
- FEBRUARY 2026 *Differential Geometry Seminar* at UC Berkeley
- JANUARY 2026 *Workshop on Connections among Spin Geometry, Minimal Surfaces and Relativity* at BIRS-CMI
- DECEMBER 2025 *Session on Geometric PDEs* at Canadian Mathematical Society Winter Meeting
- OCTOBER 2025 *Analysis and Gravitation Seminar* at Queen Mary University London
- SEPTEMBER 2025 *Geometry Seminar* at University of Waterloo
- DECEMBER 2024 *Geometric Analysis Seminar* at University of Tennessee Knoxville
- SEPTEMBER 2024 *Geometry and Topology Seminar* at University of Toronto
- JUNE 2024 *Fields Institute Geometric Analysis Colloquium*
- MAY 2024 *Informal Geometric Analysis Seminar* at Northwestern University
- APRIL 2024 *Analysis and PDE Seminar* at Notre Dame University
- MARCH 2024 *Geometric Analysis Seminar* at CUNY
- FEBRUARY 2024 *Geometry and Topology Seminar* at University of Pennsylvania
- JANUARY 2024 *Geometry Seminar* at University of Copenhagen
- DECEMBER 2023 *Analysis Seminar* at ETH
- DECEMBER 2023 *Differential Geometry Seminar* at UCSD
- DECEMBER 2023 *Geometry and Topology Seminar* at Caltech
- NOVEMBER 2023 *Geometric Analysis Seminar* at University of Chicago
- NOVEMBER 2023 *Geometric Analysis Seminar* at MIT

NOVEMBER 2023 *Recent advances in geometric analysis at CIRM*
OCTOBER 2023 *Geometric Analysis Seminar at Rutgers University*
OCTOBER 2023 *Geometry Seminar at Rice University*
SEPTEMBER 2023 *Analysis and Geometric Analysis Seminar at Cornell University*
APRIL 2023 *Graduate Student Seminar at Princeton University*

Service

REFEREE FOR: *Journal of the European Mathematical Society, International Mathematics Research Notices, Journal of Differential Geometry, Calculus of Variations and PDEs, Transactions of the American Mathematical Society.*

References

Costante Bellettini

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Robert Haslhofer

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Michael Scott Cavers (teaching)

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